Applied Psychology

Session 1

Overview

20% participation, 20% online discussion, 20% quizzes, 40% reflective journal (1000~1200 words)

Wk 2 Learning behavior. (What can you do to modify your own and other's behavior)

Wk 3 Social phychology: other people's behavior, bias. (Looking around your world: Is it the person or situation)

Wk 4 Personality. (Understanding yourself)

Wk 5 Emotion and motivation; stress, lifestyle and health. (Managing your emotions)

(Wk 6 reading week)

Wk 7 Thinking and intelligence

Wk 8 Sensation and perception

Wk 9 Lifespan development (Are adults prisoner of child hoods)

Wk 10 Biopsychology (simulating your brain: a tour)

Wk 11 Psychological disorders, therapies and treatment. (Improving your mental health: how do you distinguish between normal and abnormal behaviors)

Wk 12 Special topics in psychology

What is Psychology

Description of the discipline: Behaviour vs Mental Processes (Cognitive psychology)

To an individual: The way people make decisions, why people fell emotions, and how people interact with others

Myths correction: Psychologists **cannot** read people's mind, using **evidence** based rather than pure behavior based

Types of psychologists: educational, clinical, industrial/organizational (enhance creativity productivity), counselling psychologist

Clinical Psychology (broad field, PhD, can't prescribe med) vs Psychiatry (Doctor of Med, medical treatment of beh/mental problems)

Major Perspectives

Structuralism to evolutionary psychology

A brief history of psychology: from plato to pavlov (https://www.youtube.com/watch?v=_qqwq8EmTY4)

- 1879 Wundt: from speculation to schience, first psychology lab
- Mind: Structuralism (basic building blocks of consciousness), vs Fucntionalism (purpose of consciousness)
- Psychoanalytic by Floid: unconscious mind, techniques of free association and dream analysis
- Behaviorism: Watson & the S-R approach, focus exclusively on observable, measurable behavior, strong advocate of nurture over nature
- Behaviorism: B.F. Skinner & operant conditioning, skinner's box, strengthen and repeat positive outcomes and weaken the negative outcomes (≈learning)

Humanistic perspective: freedom & agency (free will, e.g. Sleep late isn't because mobile phone grabs you), self-actualization, positive growth

Cognitive psychology: 1950~60s internal mental processes, adopt metaphor of mind as info-processing system

Biological perspective: physiological mechanisms (e.g., dopamine is why we are so addicted to social media)

Evolutionary perspective: psychological traits and behavioral tendencies (mating preferences)

Applying Multiple Perspectives: Depression

Behavioral: learned helplessness, lack of positive reinforcement

Biological: imbalance in neurotransmitters such as serotonin and dopamine, genetic vulnerability

Cognitive: Channel way, negative automatic thoughts, distorted pattern of thinking

Psychoanalysis: trauma

Humanistic: blocked self-actualization, discrepancy between actual self and ideal self

Research Methods

Descriptive, prediction (correlational research), explanation, application

Descriptive:

case studies (especially very rare, in depth, most used by clinicians and acad researchers),

observational studies (without intervening or manipulating, naturalistic vs lab observation, e.g., parents and child in one-way mirror),

psychological test (standardized, reliable, valid measure of traits, abilities and states, objective or projective),

surveys

Correlational Research:

+ve/-ve correlation, may not be cause-effect relation!

Intervention method: independent variable (IV) vs dependent variable (DV): can prove cause-effect relation. E.g., ask experimental group to eat ice cream (IV) but not to control group => exp group have higher happy score (DV)

Exp group: receive the treatment

Control Group: do not receive the treatment (actually give conventional treatment for ethic issue)

Placebo group: inactive treatment (give fake treatment to identify and remove placebo effect)

Random assignment: eliminate systematic bias, strengthens causal claim, != random sampling

Research terminology

Hypothesis: a testable prediction about how the world will behave if our idea is correct, with scope

Operational definition: how a concept will be measured in a particular study (e.g., punishment: 2 mA electric shock)

Selecting participants: representative sample, informed consent

Session 2 Behaviorism - Learning behavior

Instincts vs Learning

Instincts is Innate, genetically programmed, inflexible; Learning is through experience, flexible responses

Classical Conditioning: reflexes associated with new stimuli

Before conditioning:

Unconditioned Stimulus (UCS) => instinctual innate response (unconditioned response UCR),

Neutral stimulus (NS) => no response

Acquisition: UCS + **association** with NS **repetitively** => UCR.

After conditioning: NS become conditioned stimulus (CS) leading to => conditioned response (CR)

E.g., be afraid of boss (UCS/UCR), a dog always accompanies boss's presence (NS) => be afraid of the dog (CS/CR)

Outcome:

generalization (CR can also be triggered by similar CS) / discrimination

Extinction (gradual weakening CR when CS repeatedly presented without UCS) & Spontaneous Recovery (fast recovery of CS by re-stimulus after extinction) & re-extinction

Operant Conditioning: Consequences shapes Behavior

Reinforcement: +Positive, - unwanted; Punishment: - Positive, + unwanted (direct reinforcement)

Self-reinforcement: Individual conduct self-reward/punishment according to one's goal

Observational / Social Learning

Vicarious reinforcement 替代强化: Observe others (role model) get Reinforcement (+ Positive, - unwanted) / Punishment, imitate /avoid their behavior

Social Cognitive Theory

Self-efficacy 自我效能: one's belief and confidence in a certain env (even without external reward), sourcing from one's own/other's experience, persuasion, emotion and physiology status.

E.g., Bobo Doll Experiment

Session 3 Social phychology: intrapersonal and interpersonal

Social cognition & attitudes

Attribution Theory 归因理论: explain behavior

Internal attribution 内部归因: inference why one's behaving in some way based something about **person**

External attribution 外部归因: inference why one's behaving in some way based something about **situation**

Attribution Theory (Examples and What it is) YT: WoTBPwMeAyo

Fundamental attribution error 基本归因谬误 (FAE): other people tend to emphasize one's internal but underestimate external factors / attribution

Actor-Observer Bias 行动者-观察者偏差

Self-service Bias 自利性偏差: one's tendency to attribute one's successes to internal, dispositional factors and explain failures by blaming external situational factors. (But people with depression think in an opposite way)

Attitudes

ABC Model: **A**ffect (fell xxx in xxx), **B**ehavior (do xxx), **C**ognition (think xxx is xxx)

Attribute formation: from Direct Contacted, Direct Instruction, Interaction with others, observational learning

Attribute Change: persuasion

Six principles:

Reciprocity (give back what you have received, Door-in-the-face: need borrow 1k$, ask 10k$ rejected -> ask 1k$ accepted),

Scarcity (Limited availability, deadline technique -> queue to buy)

Authority (following someone with credit or knowledge)

Consistency (accept small request -> accept bigger to maintain a good image, foot-in-the-door 溫水煮蛙《我係乜乜乜》(RTHK: 19/12/2015) YT:RuJOfZ_8lHg)

Liking & Ingratiation (make others favor you, offer gifts or praise)

Consensus ("75% of guests reuses towels" -> reusage rate increase 26~36%)

The science of persuasion: Who (source or communicator, credible), What (message itself, content, logic, emotional appeal "我为你着想"), What means (medium of commu, TV vs social media), Whom (target audience)

Elaboration Likelihood Model (ELM) 精细加工可能性模型:

Persuasive message -> central route / peripheral route (audience, processing, persuasion)

central route (esp. important things): Motivated & analytical, high effort in evaluating message, lasting change in attitude

peripheral route (esp. important things): not motivated nor analytical, low effort, persuaded by cues outside of message, temporary change in attitude

Cognitive dissonance 认知失调: Attitude mismatch (2 or more inconsistent attitudes/behaviors/cognitions)

Attitudes in conflict, Attitude vs. behavior conflict (e.g., want to sleep early but play phone till 3am)

Strategies to minimize cognitive dissonance: Change the behavior (quit smoking), change the cognition (The research on smoking and cancer is nonconclusive), add a new cognition (smoking suppresses my appetite, good for overall health)

Cognitive Dissonance: Your Response to Conflicting Beliefs YT: GxAu7BTZQRY

Rumor after earthquake: I am safe + I feel fear -> **Cognitive Dissonance** -> fear-justifying => those felt the tremors but suffered no damage frantically spread rumors that greater disaster would come

<u>Social Influence</u>

Compliance ≈ persuasion

Conformity 从众 (peer-pressure), e.g., Asch line experiment YT: BOBhKR4MK3w

Obedience 服从 to superior/authority figure

Milgram experiment: YT: xOYLCy5PVg: Although few people will listen to others instigation to harm others in daily life, ~65% of participants delivered highest 450V shock

Stanford/Zimbardo Prison Experiment: 24 college students – guard/prisoner group. Guard group after receiving abusing training and acting social role by wearing uniform, tend to abuse prisoner group, and many in prisoner group get depression in 2 weeks' time

Social loafing: when indi's effort is hard to measure, lead to "freerider". the larger the group, the more prominent

Deindividuation: loosening of normal behavioral constraints when individuals cannot be identified, loss of self-awareness, e.g., crowd violence, mob behavior, online trolling

Social interaction: prejudice

Stereotype (cognition, "they are xxx (bad)") => **prejudice** (emotion, "I feel xxx on them") => discrimination (beh)

Prejudice: **negative** attitude/feeling toward an indi based **solely** on membership / social group

Source: arising from realistic (group) conflict theory 现实(群体)冲突理论 (Groups become in competition for a scarcity of resources and form partiality) / social cause / psychological causes (social identity theory) => lead to in-group/out-group categorization

Pros: process new info quickly, make sense of differences between indi, enable prediction, reduce cognition effort

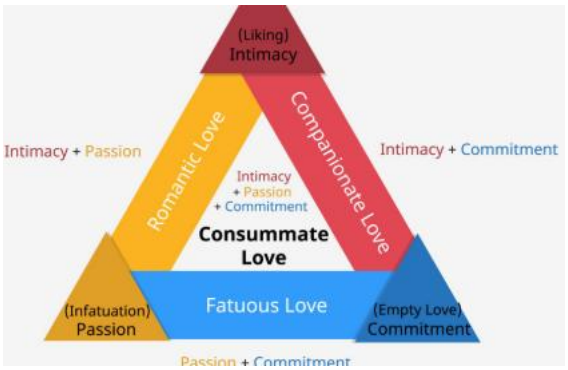
Cons: Distort reality, Selective perception, underestimates within-group diversity, in-group bias

Overcome prejudice: inter-group contact (works best), equal status

<u>Interpersonal attraction: liking and loving</u>

Driven by a fundamental human need of being affiliated

Sternberg's triangular theory of love: intimacy, passion, commitment: romantic 浪漫 (P+I), companionate 伴侣 (I+C), fatuous 愚昧 (P+C), consummate 圆满 (I+P+C, very ideal), Infatuation 迷恋 (P only)



Key factors: proximity (physical nearness), similarity (shared values and interests), physical attractiveness, self-disclosure

love yourself: self-acceptance, self-appreciation, self-understanding

Aggression & Altruism (Prosocial Behavior)

Aggression types: hostile (driven by anger), or instrumental

aggression can arise from internal or situation (e.g., frustration aggression hypothesis: hostility to factors that hinders the goal)

Pro-social behaviors (altruism): help or not help?

Factors: Notices the situation -> ambiguity of situation? current mood? perceived responsibility? Ability? helping cost? time constraint? gender of victim? physical attractiveness?

bystander intervention 旁观者干预:

bystander effect 旁观者效应 (YT:Wy6eUTLzcU4) : the more the bystanders, the less likely one will offer help,

diffusion of responsibility 责任分散: only me, then I help; in a crowd, why me not you

pluralistic ignorance 多元无知 (The smoke filled room study YT:KE5YwN4NW5o)

Session 4 Personality

IPIP-NEO-PI

Harrower-Erickson Multiple Choice Rorschach Test

http://openpsychometrics.org/tests/HEMCR

Personality: distinctive and relatively stable pattern of thoughts motives emotions behaviours,

Character (interaction) and temperament (气质 inborn)

Theoretical Approaches to Personality

Psychodynamic/psychoanalytic, Biological/Genetic, Behavioral & social-cognitive, Humanistic/Existence

Psychodynamic approach:

Who Was: Sigmund Freud | Encyclopaedia Britannica YT: cDOyT3T2RIw

Freud's core ideas: Conscious & unconscious forces, sexual drive (Eros/Libido), Aggressive drive (thanatos)

Levels of Consciousness: conscious, preconscious, unconscious

Origin of personality: result to balance two competing forces: biological drives (aggressive pleasure-seeking impulses), internal controls (internalized social rules),

ids (pleasure, desire without considering consequences) – ego (reality, strong ego=healthy personality, weak ego=psychological distress / neurosis 神经症) – superego (ideality)

Defense mechanisms (when ego overwhelmed by ids-superego conflict):

Repression: forget painful/threatening to unconsciousness,

Denial: act as if bad things does not exist,

Displacement 移情 : scapegoating, transfer bad emotion to those weaker

Rationalization: substituting socially acceptable logical reasons for one's true but less acceptable motivations

reaction formation: guarding against an unacceptable impulse by expressing its opposite

regression: reverts to coping behaviors characteristic of an earlier less mature stage, act like child,

projection: attributing one's own unacceptable thoughts/impulse to other people or the env *"it's not me, it's him!" "大环境就是这帮'强者'搞坏的"*

sublimation: redirects socially unacceptable desires or impulses into socially valued activities (the only constructing and healthy defense mech)

Psychosexual development

Freud's 5 stages of psychosexual development YT: i9o6C89MkPE

Oral (addictive overeating/undereating, smoking, alcohol, use, nail biting, excessive dependence on others for comfort and approval),

Anal: messy careless disorganized, stingy stubborn excess-neat),

Phallic: Oedipus complex (boy's) castration anxiety, electra complex (girl's) penis envy

latency, genital

limitations: neglects external factors, ignores genetic programming, lacks scientific support

Neo-Freudians: social relationships by Erik Erikson, collective unconscious by Carl Jung, inferiority and compensation by Alfred Adler, basic anxiety by Karen Horney

Biological/Genetic Approach: "fixed"

Temperament: biologically based, inborn tendencies, easy / difficult temperament

Behavioral & social-cognitive approach:

Situations & social learning: different behavior patterns across different situations, family / peer influence

Behaviorism: observable behavior, a collection of conditioned responses

Social cognitive learning by Albert Bandura: reciprocal determinism among cognitive process, context, behavior

Cultural Approach to Personality

Individualist (independence, personal achievement, and competition) vs collectivist (social harmony, respectfulness, needs of the group) cultures

Humanistic/Existence Approach

Emphasizes free will, personal agency, innate human drive toward growth

Maslow's humanistic view: (Basic) physiological -> safety needs -> (psychological) belonginess and love -> esteem -> (self-fulfillment) self-actualization

3 essential conditions for self-actualization: genuineness (openness and authentic self-disclosure: being real, not performing a role), acceptance (**unconditional** positive regard), empathy

Ideal self (person you aspire to be) vs real self (person you actually are): when ideal/real selves closely align -> **Congruence** – leads to greater self-worth, authenticity, and psychological health

Conditions of worth: conditional love -> incongruence, unconditional love -> growth

<u>Assessment of personality</u>

Self-report inventories (questionnaires where individual rate oneself on personality dimensions): MMPI 16PF NEO-PI EPQ MBTI

projective tests (open-ended stimuli, e.g., inkblots or ambiguous images, designed to reveal unconscious):

Big Five Model (OCEAN): 5 core dimension (openness, consciousness, extraversion, agreeableness, neuroticism)

Projective tests: personality assessment instruments based on freud's concept of peoject

How does the Rorschach inkblot test work YT: LYi19-Vx6go

Thematic apperception test: require the testers to tell a story about the picture

Do personality tests work? - Merve Emre YT: lN7Fmt1i5TI

Session 5 Emotion & Motivation

Theories of Emotion

Definition: a complex state of feeling that results in physical and psychological changes influencing thought and behavior

3 Components: Physiological (生理反应), Cognition (appraisal / interpretation), Behavior (facial expressions, body language, action tendencies)

James-Lange theory: stimulus -> physiological arousal -> we interpret the arousal as emotion (e.g., dog bark -> heart rate increase & run -> feel fear, rather than dog bark -> feel fear -> heart rate increase & run), start of Physiological psychology, but cannot explain why Physiological response is far slower than emotional response, same Physiological -> different emotions, physiologic impaired patient can also feel abundant emotions

Cannon-Bard theory: (reaction against James-Lange theory): physiological arousal and emotional experience occur simultaneously and independently (we tremble AND feel afraid at the same time), but didn't consider physiological in return influences emotion

Schachter-Singer two-factor theory 二因素理论: arousal 生理唤醒 (general, how strong the emotion is) + cognitive label/appraisal of the arousal (intepretation) = specific emotion. Same arousal can become different emotions depending on how we label it, e.g., rapid heart after an exam vs after a roller coaster.

Arthur Aron's suspension bridge experiment: ask two groups of men in single to cross suspension bridge or sturdy bridge respectively, and then being greeted by a beautiful female research assistant to complete TAT and call later for result interpretation.

Results show those who crossed the suspension bridge told more Sexual innuendo content in TAT and have higher calling rate. => Misattribution of arousal

Problems and Improvements: 1 is this due to sexual arousal? replace with male assistants instead => almost no calling rate; 2 is this due to those choose suspension bridge are braver and more open? Change to greeting while on the bridge / after crossing the bridge and rested for a while

Lazarus cognitive appraisal 认知评价: emotion depends on how we interpret/appraise the event (primary appraisal: is it relevant/threatening?; secondary appraisal: can I cope?)

Basic / primary emotions: happiness, sadness, anger, fear, disgust, surprise

Facial expressions are largely universal across cultures (innate, biologically wired), though display rules (how much we show them) are learned culturally

Emotional intelligence (EQ, by Goleman): ability to perceive, understand, use and manage emotions in oneself and others

4 branches: perceive -> use -> understand -> manage

EQ can be trained and predicts relationship & work success (unlike IQ)

Theories of Motivation

Motivation 动机: the process that initiates, guides and maintains goal-oriented behaviors

Origin: Needs 需要 (deficits that push us) vs Drives 驱力 (states of arousal that move us to act) vs Incentives 诱因 (external pull toward a goal)

Instinct theory 本能理论: behavior driven by innate, genetically fixed patterns (e.g., infants' sucking, maternal instinct, geese migration) (but most human behavior is learned, cannot explaining all behavior)

Drive-reduction theory 驱力降低理论 (Hull): we act to reduce internal tension/drives (hunger, thirst, cold, tiring) back to homeostasis 稳态

Arousal theory 唤醒理论: we seek an optimal level of arousal, not just reduction. Under-aroused -> seek stimulation; over-aroused -> seek calm

- Yerkes-Dodson law: performance is best at moderate arousal; too little or too much arousal impairs performance (hard tasks need low arousal, easy tasks need high)

Incentive theory 诱因理论: we are pulled by external rewards (food, money, praise) rather than pushed by internal states

Maslow's hierarchy (Session 4): physiological -> safety -> belongingness & love -> esteem -> self-actualization. Lower needs must be (mostly) satisfied before higher ones motivate behavior

Intrinsic vs Extrinsic Motivation 内在/外在动机

Intrinsic: doing something for internal satisfaction (enjoyment, curiosity). Stronger and more lasting

Extrinsic: doing something for external reward or to avoid punishment

Overjustification effect 过度合理化效应: giving external rewards for an already intrinsically rewarding task can reduce motivation (“I want to do it because it’s interesting” -> "seems I do it only for the money": Attribution change),

- Lepper, Greene and Nisbett drawing experiment: children drawing with 1 expected reward, 2 surprise reward, 3 no reward. Motivation measured: 2 >= 3 >> 1

Self-determination theory 自我决定理论 (Deci & Ryan): intrinsic motivation thrives when 3 basic psychological needs are met: autonomy 自主, competence 胜任 (≈ self-efficacy), relatedness 归属

McClelland's Learned Needs 习得性需求: achievement 成就 (desire to excel/succeed), affiliation 归属/亲和 (desire for warm social relationships), power 权力 (desire to influence/control others). High achievers prefer moderately difficult, challenging tasks

Stress, Lifestyle and Health

Stress: the process where we perceive and respond to events (stressors) that we appraise as threatening or challenging

- Primary appraisal: is this event a threat/harm/challenge to me?
- Secondary appraisal: do I have the resources/coping to deal with it?
- Eustress 良性压力 (challenge, motivating) vs Distress 恶性压力 (harmful, overwhelming)

Types of stressors: Catastrophes, Major life events (life change units, SRRS measurement), Daily hassles (often matter more for daily stress than big or rare events)

Physiology of Stress: General Adaptation Syndrome (GAS) 一般适应综合征 (Selye)

- Alarm: fight-or-flight response, sympathetic nervous system releases adrenaline/noradrenaline, mobilizes energy
- Resistance: body adapts, resists the stressor, cortisol 皮质醇 stays elevated
- Exhaustion (long-term): resources depleted, resistance collapses -> illness, burnout
- Prolonged stress = HPA axis (hypothalamus-pituitary-adrenal) overactivity, chronic cortisol -> weakened immune system (psychoneuroimmunology 心理神经免疫学), high blood pressure, heart disease, insomnia, depression

Stress & Health

- Type A personality (competitive, impatient, time-urgent, hostile, higher risk of cardiovascular disease) vs Type B (relaxed, easygoing). Hostility is the strongest predictor
- Stress suppresses the immune system: more colds, slower wound healing
- Illness/stress-buffering: strong social support (friends, family, community) reduces the harmful effects of stress

Coping

- Problem-focused coping 问题导向应对: actively change/reduce the stressor itself (make a plan, solve the problem), best when the stressor is controllable

Emotion-focused coping 情绪导向应对: manage the emotional distress (seek support, relax, reframe, avoid)

Defense mechanisms (consuming/denying) vs active coping (confronting)

Self-efficacy boosts coping; feeling in control reduces perceived stress

Lifestyle & Managing emotions (stress management)

Exercise: releases endorphins, reduces cortisol, improves mood & sleep

Sleep: sleep deprivation raises cortisol & impairs emotion regulation (sleep-deprived people overreact emotionally)

Diet: balanced diet, avoid excess caffeine/alcohol which worsen stress

Relaxation techniques: deep breathing, progressive muscle relaxation, mindfulness meditation

Social support & talking it out

Cognitive reappraisal 认知重评 (reframing a situation positively), one of the healthiest strategies

Time management, realistic goal setting, boundaries

Session 7 Thinking and intelligence

Language development: 0~3M Reflexive communication, 3~8M Reflexive communication, interest in others, 8~13 M intentional communication, sociability, 12~18 M first word

Thinking

Trial and Error: Mechanical solution, learning in puzzle box and the law of effect, exhaust solution space, systematic and require large mental capacity

Algorithm and logic: step by step procedure for trying all possible alternatives in searching for solution, Deductive / inductive reasoning YT: yAjkQ1YqLEE

Heuristics: mental shortcuts, common when facing too much info, limited time, or low-stakes decisions, by representativeness, availability, anchoring & adjustment, YT: ReFqFPJHLhA

Representativeness: judging by similarity to typical case or prototype. e.g., follow on past experience

Availability: make judgements based on how easily specific kinds of info come to mind. e.g., airplane vs car

anchoring & adjustment: tendency to use a known value as a starting reference point (anchor, frequently oneself) then adjust from it when making estimates or decisions under uncertainty. YT: NFiDdbquWJY

Barriers: mental set (persist in approaching a problem by what worked before but not now), functional fixedness (cannot perceive an object being used for sth other than what it was designed for), confirmation bias (tendency to focus on info that confirms existing beliefs), anchoring bias (focus on one info when making a decision or solving a problem), hindsight bias (event you just experienced was predictable), representative bias (intentional stereotype)

Intelligence

Intelligence: Ability to learn from experiences, acquire knowledge and use resources effectively to adapt to new situations or problem solving to survive. YT: ck4RGeoHFko

Spearman's g (general) and s (specific) factor: superiority in one type of intelligence predicts superiority overall

Cattell's Fluid (ability to see complex relationships and novel info) & Crystallized (ability to acquire and retrieve stored knowledge) intelligence

Sternberg's triarchic theory: analytical, creative, practical

Gardner's Multiple Intelligences: Verbal/linguistic intelligence, musical, logical/mathematical, visual/spatial, movement, intra/interpersonal, naturalist, existentialist intelligence

Intelligence Testing: measure general mental ability, standardization, norms, reliability & validity

Measure scales: SB-V, Wechsler (WAIS, WAIS, WPPSI)

Alfred Binet: developed in 1905, assess abstract reasoning skills, "mental age"

Lewis Terman: Intelligence Quotient (IQ) = (Mental age / chronological age) * 100

David Wechsler's scales: less reliance on verbal ability, normal distribution

Biases in intelligence testing: language bias, cultural bias, stereotype threat, ecological validity. YT: W2bKaw2AJxs

Extremes of intelligence:

intellectual disability (not mental retardation anymore): Conceptual, social, practical domain. 85% mild, 10% moderate, 5% severe, <1% profound

Causes: down's syndrome 唐氏综合征, fetal alcohol syndrome, postnatal accidents, unhealthy living conditions

Giftedness: IQ above 130 (top ~2% of the population)

Session 8 Sensation and perception

YT: lxB9bA3r5yk

Sensation: when sensory receptor detect sensory stimuli

Perception: brain orgainizes and interprets sensory info meaningfully

Sensation

Absolute thresholds: The smallest quantity human can notice, Vision, Smell, touch, taste

Difference threshold: min difference in stimuli required to detect a change or a difference between stimuli. Weber's law 2% YT: -1e0uIS5j2Q

Vision: only a small spectrum visible. What Happens Inside Your Eyes - 3D Animation YT:eySkNWTI03Q

Light -> cornea -> pupil -> fovea 中央凹(part of retina). -> rod and cones -> signals -> bipolar cells 双极细胞 -> ganglion cells 神经节细胞 -> optic nerve 视神经 -> optic chiasm 视交叉-> opposite half brain

Blind Spot. Rods (nighttime and dim light) and cones (phototopic, high-acuity color info),

Color Recognition:

Trichromatic theory (Young and Helmholtz), RGB

Opponent Process theory (Hering): RG BY BW as 3 pairs

Hearing: Loudness, Pitch, Timbre.

The science of hearing - Douglas L. Oliver YT: LkGOGzpbrCk

Outer (external) ear – middle ear (bones) – inner (internal) ear (cochlea -> auditory cortex)

Taste: papillae, taste buds. YT: wGXoYippog8

Pathway: tast buds -> impulse -> microvilli 微绒毛-> nucleus of the solitary tract -> thalamus 丘脑 -> gustatory cortex

5 basic tastes: unami salty sweet sour bitter, different people have different tastes based on genetic, culture, preference

Smell: Odor -> Olfactory 嗅觉 receptor cells (-> signal) -> olfactory bulb -> limbic system -> olfactory cortex

Sense: body position and movement of body parts: equilibrium (balance), semicircular canals

Perception

Bottom-up processing (features of stimulus -> match with schema) vs top down processing (existing schema -> break down into features -> match features with stimulus)

Gestalt principle: brain creates a perception more than sum of available sensory inputs, and brain does this in predictable ways which Gestalt psychologists translated into principles by which we organize sensory info. Figure-ground relationship (figure – focus, ground - background), proximity (tendency of grouping by nearness), similarity (tendency of grouping things alike), YT: dk7cXdjX2Ys

Theoretical explanations for perception: learning based inference (the nurture of perception), perceptual set (readiness to detect a particular stimulus in a given context)

Depth perception: Babies on the Brink YT: WanGt1G6ScA

Binocular cues: 2 eyes

Convergence: the closer the object, the greater the convergence

Retinal disparity: increases as distance between 2 object increases

Monocular Cues: linear perspective, overlapping, texture gradient, relative size, YT: Qi-nHODlQwo

Interposition: object that partially blocks second object must be closer to viewer

Linear perspective: 2 lines know parallel appear to be coming together or converging (e.g., rails)

Relative size: appear to be smaller -> farther away

Motion parallax: closer objects appear to move more quickly when viewer is in motion (e.g., on a bus)

Visual constancies: see an object as having a constant size even when its retinal images become smaller or larger

Shape constancy: continue to perceive an object as having a constant shape even though the shape of the retinal image produced by the object changes when our point of view changes.

Brightness constancy, Color constancy

Mind Illusion: 10 Mind Blowing Optical Illusions YT:-IWk5NkxQF8 How optical illusions trick your brain - Nathan S. Jacobs: YT: rfdJyDfIHIc

Optical illusions (e.g., Ponzo illusion, Muller-Lyer illusion),

Jane has prepared three cups of coffee but can't recall how much sugar is in each. The cup with the smallest amount of sugar is easy to identify, but Jane can't taste any difference between the other two cups even though she knows one has more sugar. This describes which of the following concepts: Distribution of receptors / Absolute threshold / Difference threshold?

Session 9 Lifespan development

Physical, Cognitive, Psychosocial Development

Nature & Nurture Heritability: .4~.6 personality, .8 intelligence, psychiatric illness (schizophrenia .8, alcoholism .5~.6), .3 psychosocial

Longitudinal (same person observed at different ages, hard) vs cross-sectional study (different ages observed at single time point)

Physical development

Reflexes: help baby to adapt environment (newborn reflex YT: 0vrdkzOnGgo: root, suck, tonic neck fencing, grasp, step/walking, moro)

Sensation: touch at birth, smell, taste, hearing 0~1M vision cones (~6M depth 4~5M, mother face recognition ~2W)

Habituation: YT: dlilZh60qdA

Brain size: 24M 55% of adult size, 6Y 90% of adult, development YT: ZeEKLECvh9g with synaptic pruning 突触修剪 YT: rxPT78F_ZVE

Motor skills (fine/gross): ability to move bodies and manipulate objects

Mother-child interaction

Parenting styles: dimension of demandingness / responsiveness (=> authoritative +R +D, authoritarian -R +D, permissive +R -D, neglectful -R -D) + over-involved/helicopter parenting YT: fyO8pvpnTdE

Temperament (easy/difficult/slow-to-warm-up): innate traits that influence how one thinks behaves and reacts with environment

Attachment: a long-standing connection or bond with others

The Attachment Theory: How Childhood Affects Life YT: WjOowWxOXCg

Bowlby & Ainsworth: importance of contact comfort in attachment YT: -Qi7txH1KzY / awitQqurqEI

Strange situation: procedure to study attachment between mothers and their infants, Mary Ainsworth's study YT: m_6rQk7jlrc secure/ambivalent/avoidant/disorganized attachment

https://novopsych.com/assessments/formulation/attachment-style-questionnaire-short-form-asq-sf/

Cognitive development

Piaget's stage theory: sensorimotor 0~2 years, preoperational 2~7Y, concrete operation, complete

Schemata: concepts that are used to help us categorize and interpret information, adjust through assimilation and accommodation

Sensorimotor: object permanence ~8M (out of sight existence YT: cSGWh2CWJnA)

Preoperational stage: representing ideas with words and images but lacking logical reasoning, achieving egocentrism (YT: RDJ0qJTLohM 3 mountain experiment), animistic thinking (believe inanimate objects have life and mental processes), centration (inability to think more than 1 angle YT: GLj0IZFLKvg), irreversibility (cannot think back, inability to think through a series of events or mental operation and reverse the steps, lacks mental trial and error ability of do and redo)

Concrete Operational ~8~12Y, understands conservation and incapable of abstract thought YT: QxUxgPwpfgk

Formal Operational ~12+Y: ability for abstract and complex though, think hypothetical questions (love, justice, freewill, reasons of existence), further developments in thinking are changes in degree rather than fundamental changes in the nature of thinking, teens use systematic thinking strategies to solve problems with trial-and-error approach

Language Development: Cooing 0~3M, Babbling 6~12M, one-word 12~18M, telegraphic speech 18~24M, whole sentences 36~72M, Critical period for language development YT: 9iAxfgVkqDU

Psychosocial

Erikson's 8 stages of psychosocial development explained YT:aYCBdZLCDBQ, basic trust vs mistrust at infant ~1Y, autonomy vs shame and doubt at toddler 1~3Y, initiative vs guilt at preschool 3~5Y, Industry vs inferiority at elementary school 5~12Y, identity vs role confusion at adolescence 13~19Y, intimacy vs isolation at early adulthood (20~40Y), generativity vs stagnation at middle adulthood (40~65Y), ego integrity vs despair at late adulthood (65~death)

Moral development: Heinz Dilemma YT: 5czp9S4u26M

Kohlberg's stages of morality: preconventional (based on consequences that avoid punishment and bring rewards) -> conventional (based on intentions norms and rules/laws) -> postconventional (based on mutual benefit and utilitarianism, inner conscience and ethical actions as ends not means, universal principles)

Session 10 Biopsychology

Session 11 Psychological disorders, therapies and treatment

Session 12 Special topics in psychology